

LUMBINI ENGINEERING COLLEGE

Internal examination

Year: 2024

Full Marks:100

Pass Marks:45

Time :3hrs.

Level: Bachelor

Programme: BE (Computer)

Course: Calculus II

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

6x15=90

Attempt all the questions.

1a) Evaluate the given integral.

$$\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2 + y^2) dx dy \text{ for } a>0, \text{ (by changing into polar form) OR}$$

$$\iiint (x^2 z^2 + z^2 y^2 + y^2 x^2) dx dy dz \text{ throughout the volume of the sphere } x^2 + y^2 + z^2 = b^2.$$

b) Find the volume bounded by xy-plane, the paraboloid $2z = x^2 + y^2$ and the cylinder $x^2 + y^2 = 4$.

2a) Solve by using power series method. $(1+x^2)y'' + xy' - y = 0$.

b) What is Legendre's equation? Find its general solution.

3a) Define Laplace transforms. State first and second shifting theorem of Laplace transform.

Find i) $L(t \cos wt)$ ii) $L^{-1}\left(\frac{se^{-2s}}{s^2+1}\right)$

b) Solve the given initial value problem by using Laplace transform.

$$y'' + 2y' - 3y = 6e^{-2t}, y(0) = 2, y'(0) = -14$$

4a) Show that the value under the integral sign is exact in the plane and evaluate

$$\int_{(0,1,1/2)}^{(\pi/2,3,2)} [y^2 \cos x dx + (2y \sin x + e^{2z}) dy + 2ye^{2z} dz]$$

b) Evaluate $\iint_S \vec{F} \cdot \vec{n} dA$, Where $\vec{F} = (xy, yz, xy)$ and S is the part of the surface $x^2 + y^2 + z^2 = 1$, which lies in the first octant.

5a) Verify stoke's theorem for the vector function $\vec{F} = (y^3, 0, x^3)$ and C is the boundary of the triangle with vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1).

b) Write the Fourier coefficient of a function $f(x)$. Find the Fourier expansion of the function $f(x) = x + |x|$ ($-\pi < x < \pi$).

6a) Find the Fourier sine as well as cosine series representation of the half range of the function $f(x) = x^2$ for $0 < x < L$.

b) Find the breaking time for $u_t + uu_x = 0$, $u(x, 0) = e^{-x^2}$.

4x2.5=10

7a) Find $f(t)$ for $\frac{2s+6}{s^2+4}$

b) Find the unit tangent vector at any point on the curve $x = 3\cos t, y = 3\sin t, z = 4t$.

c) Define even and odd function and find it for $x \cos nx$.

d) Define linear first order partial differential equation.